

PROBABILITY REVIEW

Probability Review

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$$P(A \mid B) \quad \mathbb{E}[X] \quad \text{Var}(X)$$

Probability review map

1. **What can happen?** Experiments, sample spaces, events, and set operations
2. **How likely is it?** Probability rules, conditioning, independence, and Bayes' rule
3. **How do numbers inherit uncertainty?** Random variables, distributions, mean, and variance

Guiding habit

Write the event or integration region first. Choose the formula second.

Experiments, sample spaces, and events

Definition

An **experiment** produces an outcome that cannot be predicted with certainty. The **sample space** S contains all possible outcomes. An **event** E is any subset of S .

In plain words

First list outcomes at the level of detail recorded by the experiment. An event then selects the outcomes that satisfy a condition.

Experiments, sample spaces, and events

Practice Problem

Toss two coins and record the side on each coin. Let H be the event *exactly one head*. Write the sample space S and the event H .

Work on the problem before advancing to the solution.

Solution: experiments and events

Solution

$$S = \{HH, HT, TH, TT\}, \quad H = \{HT, TH\}.$$

If we record only the *number* of heads, the sample space changes to $\{0, 1, 2\}$.

Operations on events

Definition

A^c contains outcomes not in A . The intersection $A \cap B$ contains outcomes in both events. The union $A \cup B$ contains outcomes in at least one event.

In plain words

Complement means *not*. Intersection means *and*. Union means the inclusive *or*.

$$(A \cap B)^c = A^c \cup B^c, \quad (A \cup B)^c = A^c \cap B^c.$$

Operations on events

Practice Problem

Let $S = \{0, 1, 2, 3, 4, 5\}$ and $A = \{0, 2, 4\}$. Find A^c . Separately, let $C = \{1, 2, 3\}$ and $B = \{2, 4, 6\}$. Find $C \cap B$ and $C \cup B$.

Work on the problem before advancing to the solution.

Solution: operations on events

Solution

$$\begin{aligned}A^c &= \{1, 3, 5\}, & C \cap B &= \{2\}, \\C \cup B &= \{1, 2, 3, 4, 6\}.\end{aligned}$$

Mutually exclusive and exhaustive events

Definition

Events are **mutually exclusive** when $A \cap B = \emptyset$. They are **exhaustive** when $A \cup B = S$. A collection that is both mutually exclusive and exhaustive forms a **partition** of S .

In plain words

Disjoint events cannot occur together. Exhaustive events cover every outcome. Complements have both properties.

Mutually exclusive and exhaustive events

Practice Problem

Roll a fair six-sided die. Let A be *even* and B be *odd*. Are A and B mutually exclusive, exhaustive, complementary, or more than one of these?

Work on the problem before advancing to the solution.

Solution: mutually exclusive and exhaustive events

Solution

$$A = \{2, 4, 6\}, \quad B = \{1, 3, 5\}.$$

$A \cap B = \emptyset$, so they are mutually exclusive. $A \cup B = S$, so they are exhaustive. Since $A = B^c$, they are complementary and form a partition.

Probability axioms and equally likely outcomes

Definition

A probability model satisfies

1. $P(S) = 1$,
2. $P(A) \geq 0$,
3. Probabilities add across mutually exclusive events. If $A_1, A_2, \dots$, are mutually exclusive events, then $P(A_1 \cup A_2 \cup \dots) = P(A_1) + P(A_2) + \dots$

If S has N equally likely outcomes and A contains k of them, then $P(A) = k/N$.

In plain words

Counting works only after checking that the elementary outcomes are equally likely.

Probability axioms and equally likely outcomes

Practice Problem

Roll a fair six-sided die once. Find the probability of rolling a factor of 8. State the favorable outcomes before calculating.

Work on the problem before advancing to the solution.

Solution: equally likely outcomes

Solution

The factors of 8 that appear on the die are $\{1, 2, 4\}$. There are three favorable outcomes among six equally likely outcomes:

$$P(A) = \frac{3}{6} = \boxed{\frac{1}{2}}.$$

Complement and addition rules

Definition

$$P(A^c) = 1 - P(A),$$
$$P(A \cup B) = P(A) + P(B) - P(A \cap B).$$

In plain words

The overlap appears in both $P(A)$ and $P(B)$, so subtract it once.

Complement and addition rules

Practice Problem

Let D denote the event that the selected student is majoring in Data Science, and let L denote the event that the student is in the College of Literature, Science, and the Arts (LSA). Suppose

$$P(D) = 0.16, \quad P(L) = 0.22, \quad P(D \cap L) = 0.10.$$

- (a) What is the probability that a randomly selected student is majoring in Data Science or is in LSA?
- (b) What is the probability that a randomly selected student is neither majoring in Data Science nor is in LSA?
- (c) What is the probability that a randomly selected student is majoring in Data Science but is not in LSA?

Work on the problem before advancing to the solution.

Solution: complement and addition rules

Solution

$$P(D \cup L) = 0.16 + 0.22 - 0.10 = \boxed{0.28},$$

$$P(\text{neither}) = 1 - 0.28 = \boxed{0.72},$$

$$P(D \cap L^c) = 0.16 - 0.10 = \boxed{0.06}.$$

Conditional probability

Definition

$$P(A \mid B) = \frac{P(A \cap B)}{P(B)}.$$

Once B is known, only outcomes inside B remain possible. The condition changes the reference group.

Conditional probability

Practice Problem

A wafer is selected uniformly from this table.

Lot	Conforming	Nonconforming	Total
A	88	12	100
B	165	35	200
C	260	40	300
Total	513	87	600

Find $P(B \cap G)$, $P(B | G)$, and $P(G | B)$, where B means Lot B and G means conforming.

Work on the problem before advancing to the solution.

Solution: conditional probability

Solution

$$P(B \cap G) = \frac{165}{600} = \boxed{0.275},$$

$$P(B \mid G) = \frac{165}{513} = \boxed{0.3216},$$

$$P(G \mid B) = \frac{165}{200} = \boxed{0.825}.$$

The denominators differ because each condition creates a different reference group.

Independence of events

Definition

A and B are **independent** when knowing B does not change the probability of A .

$$P(A \mid B) = P(A), \quad P(A \cap B) = P(A)P(B).$$

In plain words

Mutually exclusive events with positive probability are dependent. If one occurs, the other becomes impossible.

Independence of events

Practice Problem

Use the wafer data below.

Lot	Conforming	Nonconforming	Total
A	88	12	100
B	165	35	200
C	260	40	300
Total	513	87	600

Determine whether *Lot B* and *conforming* are independent. Show the two probabilities that justify your answer.

Work on the problem before advancing to the solution.

Solution: independence of events

Solution

$$P(B) = \frac{200}{600} = 0.3333, \quad P(B \mid G) = \frac{165}{513} = 0.3216.$$

The probabilities differ, so the events are dependent under this model.

Law of total probability and Bayes' rule

Law of total probability

If $A_1, A_2, \dots, A_n$ are mutually exclusive and exhaustive events, and B is any event, then

$$P(B) = P(A_1 \cap B) + P(A_2 \cap B) + \dots + P(A_n \cap B).$$

Equivalently,

$$P(B) = P(A_1)P(B | A_1) + \dots + P(A_n)P(B | A_n) = \sum_{i=1}^n P(A_i)P(B | A_i).$$

Bayes' rule

Let $A_1, A_2, \dots, A_n$ be mutually exclusive and exhaustive events with prior probabilities $P(A_i)$. After observing B , the posterior probability of A_j is

$$P(A_j | B) = \frac{P(A_j \cap B)}{P(B)} = \frac{P(A_j)P(B | A_j)}{\sum_{i=1}^n P(A_i)P(B | A_i)}.$$

Prior: $P(A_i)$ before observing B . **Posterior:** $P(A_i | B)$ after observing B .

Bayes' rule updates the prior after observing B .

Practice Problem

Let D denote a bottle with a microscopic flaw and F a failed inspection. Given $P(D) = 0.0002$, $P(F | D) = 0.995$, and $P(F | D^c) = 0.01$, find $P(D | F)$.

Work on the problem before advancing to the solution.

Solution: rare defect after a failed inspection

Solution

By Bayes' rule and the law of total probability,

$$P(D | F) = \frac{P(D)P(F | D)}{P(D)P(F | D) + P(D^c)P(F | D^c)}.$$

Now substitute the given probabilities:

$$P(D \cap F) = 0.0002(0.995) = 0.000199,$$

$$P(F) = 0.000199 + 0.9998(0.01) = 0.010197,$$

$$P(D | F) = \frac{0.000199}{0.010197} \approx \boxed{0.0195}.$$

Only about 1.95% of failed bottles actually have the flaw. The rare prior makes false positives dominate the pool of failures.

Random variables

Definition

A random variable X assigns a numerical value to every outcome. It is **discrete** if its possible values form a finite or countably infinite set. It is **continuous** if its values fill an interval.

In plain words

The experiment supplies the randomness. The function X records a numerical feature of the outcome.

Practice Problem

Classify each random variable as discrete or continuous: the number of spam emails received in one hour, a transistor's lifetime, and the number of occupants in a car. Briefly justify each classification.

Work on the problem before advancing to the solution.

Solution: classifying random variables

Solution

- Spam count: discrete, with values $0, 1, 2, \dots$
- Transistor lifetime: continuous, with values in an interval such as $(0, \infty)$
- Occupant count: discrete, with positive integer values

A measuring device may round a continuous quantity, but the probability model can remain continuous.

Probability mass functions

Definition

For a discrete random variable,

$$p(x) = P(X = x).$$

A valid pmf satisfies $p(x) \geq 0$ and $\sum_x p(x) = 1$. For a set A ,

$$P(X \in A) = \sum_{x \in A} p(x).$$

Probability mass functions

Practice Problem

The number of spam emails in one hour has pmf

x	0	1	2	3	4
$p(x)$	.81	.13	.04	.01	.01

Find $P(X > 3)$, $P(X \geq 2)$, and $P(X = 3 \mid X \geq 2)$.

Work on the problem before advancing to the solution.

Solution: probability mass functions

Solution

$$P(X > 3) = \boxed{0.01},$$

$$P(X \geq 2) = 0.04 + 0.01 + 0.01 = \boxed{0.06},$$

$$P(X = 3 \mid X \geq 2) = \frac{0.01}{0.06} = \boxed{\frac{1}{6}}.$$

Cumulative distribution functions

Definition

$$F(x) = P(X \leq x).$$

For a discrete variable, F is a step function. The jump at a possible value x equals $p(x)$.

In plain words

The cdf accumulates probability from the left. It is nondecreasing and approaches 1 as x increases.

Cumulative distribution functions

Practice Problem

A car has 1, 2, 3, 4, or 5 occupants with probabilities .70, .15, .10, .03, .02. Construct the cdf $F(x)$ and find $P(X \leq 2.5)$.

Work on the problem before advancing to the solution.

Solution: cumulative distribution functions

Solution

$$F(x) = \begin{cases} 0, & x < 1, \\ .70, & 1 \leq x < 2, \\ .85, & 2 \leq x < 3, \\ .95, & 3 \leq x < 4, \\ .98, & 4 \leq x < 5, \\ 1, & x \geq 5. \end{cases}$$

$$P(X \leq 2.5) = p(1) + p(2) = 0.85 = F(2.5).$$

Mean, variance, and standard deviation

Center

$$\mathbb{E}[X] = \mu = \sum_x xp(x).$$

The mean is the probability-weighted long-run average. It need not be a possible value of X .

Spread

$$\text{Var}(X) = \sum_x (x - \mu)^2 p(x)$$

$$= \mathbb{E}[X^2] - (\mathbb{E}[X])^2.$$

The standard deviation $\text{SD}(X) = \sqrt{\text{Var}(X)}$ uses the original units.

Solvent drum orders

Practice Problem

The number of drums ordered has distribution

x	1	2	3	4	5
$p(x)$	.10	.10	.60	.15	.05

Find $E[X]$, $\text{Var}(X)$, $\text{SD}(X)$, and the probability that X lies within two standard deviations of its mean.

Work on the problem before advancing to the solution.

Solution: solvent drum orders

Formulas

$$\mathbb{E}[X] = \sum_x xp(x), \quad \mathbb{E}[X^2] = \sum_x x^2 p(x), \quad \text{Var}(X) = \mathbb{E}[X^2] - (\mathbb{E}[X])^2, \quad \text{SD}(X) = \sqrt{\text{Var}(X)}.$$

Substitute into the moment formulas

$$\begin{aligned}\mathbb{E}[X] &= 1(.10) + 2(.10) + 3(.60) + 4(.15) + 5(.05) \\ &= 2.95,\end{aligned}$$

$$\begin{aligned}\mathbb{E}[X^2] &= 1^2(.10) + 2^2(.10) + 3^2(.60) + 4^2(.15) + 5^2(.05) \\ &= 9.55,\end{aligned}$$

$$\text{Var}(X) = 9.55 - 2.95^2 = 0.8475,$$

$$\text{SD}(X) = \sqrt{0.8475} \approx 0.921.$$

Within two standard deviations

$$P(|X - \mu| \leq 2\sigma) = P(\mu - 2\sigma \leq X \leq \mu + 2\sigma).$$

$$2.95 \pm 2(0.921) \text{ gives } [1.108, 4.792].$$

The possible values are 2, 3, and 4, so

$$P(2 \leq X \leq 4) = .10 + .60 + .15 = \boxed{.85}.$$

Continuous random variables and density

Definition

A probability density function $f(x)$ is nonnegative and has total area 1:

$$\int_{-\infty}^{\infty} f(x) dx = 1.$$

Interval probabilities are areas:

$$P(a < X < b) = \int_a^b f(x) dx.$$

In plain words

A density value is a height, not a probability. For a continuous variable, $P(X = c) = 0$, so changing $<$ to $\leq$ does not change an interval probability.

Continuous random variables and density

Practice Problem

Can a probability density exceed 1? If so, give a valid continuous density whose height is greater than 1 and verify that its total area is 1.

Work on the problem before advancing to the solution.

Solution: density values greater than 1

Solution

Yes. If $X \sim \text{Uniform}(0, 0.5)$, then

$$f(x) = 2, \quad 0 < x < 0.5.$$

The density height is greater than 1, but its total area is

$$2(0.5) = \boxed{1}.$$

The value $f(x) = 2$ is a **density height**, not the probability $P(X = x)$. In fact, $P(X = x) = 0$ for every single value of a continuous random variable. Probabilities are areas under the density:

$$P(a < X < b) = \int_a^b f(x) dx = 2(b - a) \leq 1 \quad (0 \leq a < b \leq 0.5).$$

Therefore the pdf height may exceed 1, but a probability can never exceed 1.

The relationship between the cdf and pdf

Definition

$$F(x) = P(X \leq x) = \int_{-\infty}^x f(t) dt.$$

Where F is differentiable,

$$f(x) = F'(x).$$

In plain words

Integrating density accumulates probability from the left. Differentiating the cdf recovers local density.

The relationship between the cdf and pdf

Practice Problem

Let $F(x) = 0$ for $x < 0$ and $F(x) = 1 - e^{-x^2}$ for $x \geq 0$. Find $P(X > 1)$ and derive the density $f(x)$.

Work on the problem before advancing to the solution.

Solution: recovering a pdf from a cdf

Solution

$$P(X > 1) = 1 - F(1) = e^{-1} \approx \boxed{0.3679}.$$

For $x > 0$,

$$f(x) = \frac{d}{dx}(1 - e^{-x^2}) = \boxed{2xe^{-x^2}}.$$

For $x \leq 0$, $f(x) = 0$.

Mean and variance for a continuous variable

Definition

$$\mathbb{E}[X] = \int_{-\infty}^{\infty} x f(x) dx,$$

$$\text{Var}(X) = \int_{-\infty}^{\infty} (x - \mu)^2 f(x) dx = \mathbb{E}[X^2] - (\mathbb{E}[X])^2.$$

Mean and variance for a continuous variable

Practice Problem

Snow depth has density $f(x) = 9xe^{-3x}$ for $x > 0$. Find $E[X]$, $\text{Var}(X)$, and $\text{SD}(X)$.

Work on the problem before advancing to the solution.

Solution: continuous mean and variance

Solution

Direct integration gives

$$\mathbb{E}[X] = \frac{2}{3} \approx 0.667 \text{ inches},$$

$$\text{Var}(X) = \frac{2}{9}, \quad \text{SD}(X) = \frac{\sqrt{2}}{3} \approx \boxed{0.471 \text{ inches}}.$$

Medians and percentiles

Definition

The 100 p th percentile x_p solves

$$F(x_p) = p.$$

The median is $x_{0.50}$. The interquartile range is

$$\text{IQR} = x_{0.75} - x_{0.25}.$$

In plain words

Percentiles invert the cdf: choose a cumulative probability and solve for the corresponding value.

Medians and percentiles

Practice Problem

For $F(x) = 1 - e^{-0.1x}$, find the median, first quartile, third quartile, and interquartile range.

Work on the problem before advancing to the solution.

Solution: exponential percentiles

Solution

Solving $1 - e^{-0.1x_p} = p$ gives

$$x_p = -\frac{\ln(1-p)}{0.1}.$$

$$x_{0.50} = 6.931, \quad x_{0.25} = 2.877, \quad x_{0.75} = 13.863,$$

$$\text{IQR} = 13.863 - 2.877 = \boxed{10.986 \text{ months}}.$$